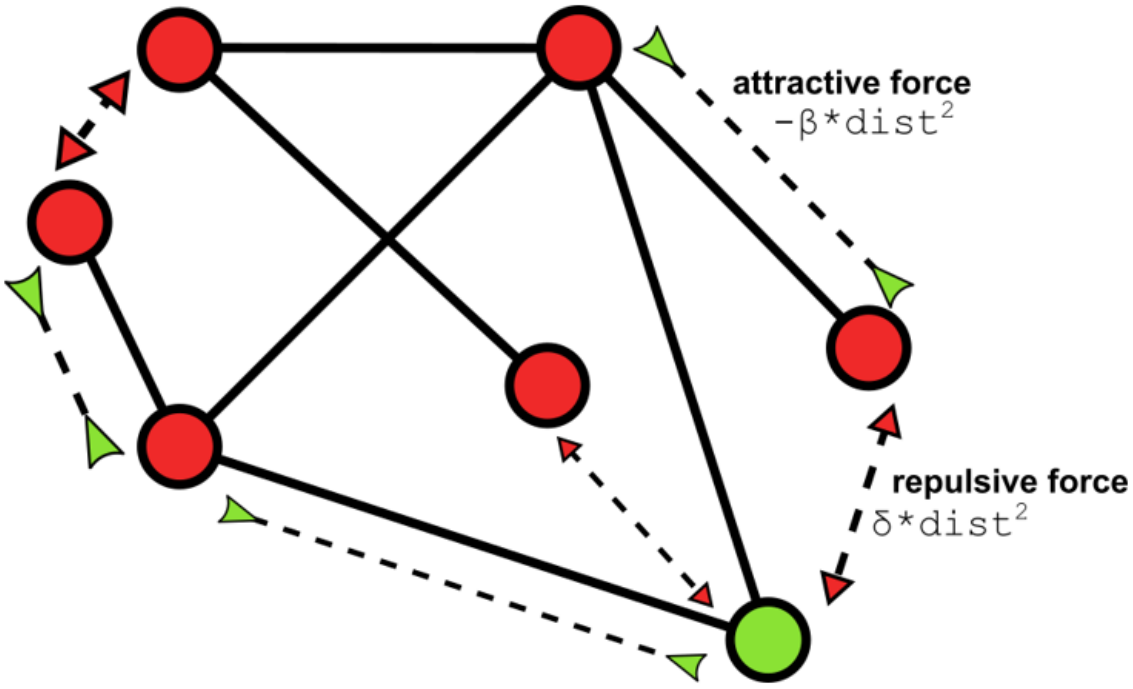


Figure 16.20 A force-directed drawing algorithm using physics simulation, with attractive and repulsive forces acting on vertices, to provide an aesthetically pleasing embedding. Notice how forces are larger (thicker lines) when vertices are closer. For clarity sake, we don't show all the forces acting on all pairs of vertices, but just a few examples.

To further refine our initial analogy, we can use springs (or gravity) as the physics counterpart of edges, and a fainter electrical repulsion[19] among all pair of vertices. Note that all these forces depend on the distance between the vertices – you can replace them with different formulas, as long as you keep this characteristic. Figure 16.20 gives you an idea of how such systems can work.

The next thing we need is formalizing these criteria into a formula for the cost function: that will describe the landscape of the problem, which we can then try to explore by using gradient